Landslide Monitoring and Risk — Colombia

Workshop No. 1 — Hazard Definition, Source Validation, and Data Contract
Data Analysis Programming, Semester 2026-II — Universidad Distrital Francisco José de Caldas
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Repository: `https://github.com/SARA_USER/landslide-monitoring-colombia` — Sources queried on 7 November 2026 (UTC)

1. Project Overview

Title. Landslide Monitoring and Risk — Colombia.

Motivation. Rainfall-triggered landslides are one of the most frequent and deadly natural hazards in Colombia. The Andean departments combine steep slopes, deep tropical soils and two rainy seasons per year; the Salgar (2015) and Mocoa (2017) disasters alone killed more than 400 people according to the NASA catalog (92 and 329 deaths). The information needed to watch this hazard exists, but it is spread over sources with different formats, units, identifiers and update rhythms: an inventory of past landslides, a rainfall reanalysis, a rainfall forecast service and a satellite-based nowcast.

Goal. From a data-engineering perspective, build a reproducible pipeline that ingests two historical and two periodically updated sources into one canonical model, detecting new and changed records with watermarks. From an analytical perspective, produce a *Landslide Risk Indicator per region* that tells an analyst where current rainfall and nowcast conditions resemble those that preceded past landslides.

Scope. Colombia (bounding box lat −4.3 to 13.5, lon −79.1 to −66.8); landslide history 2007–2017; rainfall history 2007–present; current conditions updated hourly (rainfall) and daily (nowcast); analysis on a 0.25° grid of analysis cells.

Explicit exclusions. No prediction of individual landslides; no earthquake-triggered or human-triggered landslide modelling; no field or geotechnical data; no countries other than Colombia; no alert dissemination to the public; no paid or registration-only sources.

Constraints. Free and open sources only; Open-Meteo free tier (non-commercial, rate-limited); everything runs locally with Docker Compose; documentation in English.

2. Hazard Definition

Phenomenon. A landslide is the downslope movement of soil, rock or debris under gravity. In Colombia most landslides are triggered by rainfall: water infiltrates the slope, raises pore-water pressure and reduces the shear strength of the soil until the slope fails. Both short intense storms and long wet periods matter, which is why *antecedent rainfall* (accumulated over several days) is the main monitoring variable.

How it is measured and its severity. Landslides have no single instrumental magnitude. This project uses three complementary measures, always stored with their unit:

Measure	Source	Unit / scale	Meaning
Landslide size and impact	NASA Global Landslide Catalog	Size class 1–5 (small … catastrophic); fatalities (persons)	Severity of a past event
Triggering rainfall	Open-Meteo (ERA5 and forecast)	mm/day, mm/h; accumulations over 1, 3, 7 and 14 days (mm)	Physical driver of the hazard
Nowcast hazard	NASA LHASA v2	Dimensionless value 0–1 per ~1 km pixel	Model estimate of current landslide hazard

Why monitoring matters. Rainfall and slope conditions change over days, so a timely picture of where antecedent rainfall is unusual helps risk managers decide where to inspect roads and slopes and where to prepare communities.

Geographic scope and time window. Colombia, because it is relevant to the team and is covered by all four sources (all are global). Landslide history is limited to **2007–2017** because the NASA catalog stops in September 2017 (107 Colombian events, 2007-05-21 to 2017-04-19). Rainfall history covers **2007–present** (about 20 years), which gives a stable climatology for comparing current rainfall. The bulk load is affordable: a 0.25° grid over the terrain evaluated by LHASA is 782 cells × 7,250 days ≈ 5.7 million daily rows (Section 6.5).

Current situation. For this hazard, "current" means: (1) observed rainfall in the last 24 hours and the last 7 days, (2) the forecast for the next 72 hours, and (3) the latest available LHASA nowcast (normally the previous day, because files are published about 29 hours after their nominal time).

3. Data Sources

All four sources were queried for real on 7 November 2026. The responses are kept in `workshop_1/samples/` exactly as received, with their HTTP headers and request URLs: the full catalog CSV, the Open-Meteo JSON responses, the LHASA directory listings and HEAD headers, and the LHASA raster metadata. Two sample sets are *derived* and labelled as such: the LHASA Colombia windows (clipped from the global file and flipped north-up) and the 107 antecedent-rainfall responses (each response wrapped together with its request). None of the sources requires a key.

	H1 — NASA Global Landslide Catalog	**H2 — Open-Meteo Historical**	**I1 — Open-Meteo Forecast**	**I2 — NASA LHASA nowcast**
Owner	NASA Goddard Space Flight Center	Open-Meteo (ERA5 by Copernicus/ECMWF)	Open-Meteo	NASA Goddard Space Flight Center
URL	`data.nasa.gov/docs/legacy/Global_Landslide_Catalog_Export/Global_Landslide_Catalog_Export_rows.csv`	`archive-api.open-meteo.com/v1/archive`	`api.open-meteo.com/v1/forecast`	`portal.nccs.nasa.gov/datashare/landslides/nrt/hazard/tif/`
Access	One CSV file; 302 redirect to a pre-signed S3 URL (valid 3,600 s)	REST GET, many locations per request	REST GET, many locations per request	Directory listing + one GeoTIFF per issue time; HTTP range requests
Auth	None (a descriptive User-Agent is sent)			
Rate limits	None documented	600/min, 5,000/h, 10,000/day; weight = locations × days/14 × variables/10 (confirmed by `X-RateLimit-Cost`)		None documented; ~140 MB per global file
License / attribution	U.S. Government work; cite Kirschbaum et al. (2010, 2015)	CC BY 4.0, non-commercial free tier; credit "Weather data by Open-Meteo.com", ERA5 by C3S/ECMWF		U.S. Government work; cite Kirschbaum & Stanley (2018), Stanley et al. (2020)
Format / fields	CSV, 31 columns: event_id, event_date, event_title, location_accuracy, landslide_category/trigger/size, fatality_count, country_name, latitude, longitude, …	JSON: latitude, longitude (grid), elevation, daily.time, precipitation_sum, rain_sum, precipitation_hours, *_units	JSON: current (15 min), hourly.time, precipitation, rain, soil_moisture_0_to_1cm, *_units	GeoTIFF float64, EPSG:4326, 0.00833° (~1 km), 43,200 × 14,400 px, tiled 512×512, zstd (saved metadata of `20261106T0000.tif`)
Coverage	1988-11-07 → 2017-09-28; 11,033 events, 107 in Colombia	Global reanalysis; 2007–present requested for this project	Global; 7 past days + 3 forecast days requested	Global (60°S–60°N), 2025-01-01 → 2026-11-06 in the NRT folder
Update policy	Frozen; file Last-Modified 2025-05-29	Extended daily	Model runs several times per day; current every 15 min	Daily (best effort), ~29 h delay, gaps

The two historical sources use different formats and access methods (a bulk CSV file vs. a parameterized JSON API), as required.

3.1 Evidence: real response snippets

H1 — NASA catalog (row of `nasa_glc_colombia_bbox_20261107_045852.csv`, selected columns, and the redirect header):

```
event_id=10790 | event_date=03/25/2017 05:32:00 PM | event_title=Deadly Mocoa Landslide | country_name=(empty)
location_accuracy=1km | landslide_trigger=downpour | landslide_size=catastrophic | fatality_count=329
latitude=1.151898036 | longitude=-76.63992276 | last_edited_date=02/15/2018 03:51:00 PM
HTTP/2 302 · location: https://data-nasa-bucket-production.s3.us-east-1.amazonaws.com/legacy/...csv?[pre-signed query removed]
HTTP/1.1 200 OK · Last-Modified: Thu, 29 May 2025 19:19:59 GMT · ETag: "7e12dbd01cd937d7f3a68df0f3fcecce"
```

H2 — Open-Meteo historical (Mocoa, 2016–2017 request; response truncated):

```
{"latitude": 1.0896308, "longitude": -76.59573, "utc_offset_seconds": 0, "timezone": "GMT", "elevation": 597.0, "location_id": 3,
 "daily_units": {"time": "iso8601", "precipitation_sum": "mm", "rain_sum": "mm", "precipitation_hours": "h"},
 "daily": {"time": ["2016-01-01", "2016-01-02", "2016-01-03", ...], "precipitation_sum": [1.1, 5.7, 0.0, ...], ...}}
X-RateLimit-Cost: 156.64287
```

I1 — Open-Meteo forecast (Manizales, poll of 2026-11-07 04:47:46 UTC; response truncated):

```
"current": {"time": "2026-11-07T04:45", "interval": 900, "precipitation": 0.2, "rain": 0.0}
"hourly_units": {"time": "iso8601", "precipitation": "mm", "rain": "mm", "soil_moisture_0_to_1cm": "m³/m³"}
"hourly": {"time": ["2026-11-06T06:00", "2026-11-06T07:00", "2026-11-06T08:00"], "precipitation": [0.2, 3.4, 0.9],
           "rain": [0.1, 0.1, 0.2], "soil_moisture_0_to_1cm": [0.395, 0.423, 0.421]}
```

I2 — LHASA (directory listing of 2026-11-07 05:58 UTC, last rows; HEAD of the newest file):

```
20261104T0000.tif   139.1 MiB   November 5, 2026
20261105T0000.tif   139.2 MiB   November 6, 2026
20261106T0000.tif   138.3 MiB   November 7, 2026
HEAD 20261106T0000.tif → Content-Length: 145027040 · Last-Modified: Sat, 07 Nov 2026 05:07:25 GMT · ETag: "6aeeb20d-8a4efe0"
```

3.2 Historical volume and bulk-load partitioning

Source	Full historical volume	Partitioning of the bulk load
NASA catalog	1 file, 8.5 MB, 11,033 rows (107 for Colombia)	None needed: one download; the global file is kept in the raw zone and filtered in normalization.
Open-Meteo historical	782 cells (0.25°) × 7,250 days (2007-01-01 → 2026-11-06) ≈ 5.67 M rows, about 55 MB of raw JSON (9.6 bytes per value measured in the sample) and roughly 2 GB in PostgreSQL (event plus rainfall_observation rows with indexes, about 350 bytes per observation; to be measured in Workshop #2); with one variable the request weight is ≈ 40,500, i.e. about 4 days of free quota (0.5° grid: 194 cells, ≈ 1 day)	Groups of cells × calendar year; each partition is one request and one transaction; the run stops before the daily limit and resumes from `ingestion_state`.

4. Incremental Behaviour (empirical)

Both incremental sources were polled three times on 7 November 2026. Open-Meteo forecast: 04:47:46, 05:58:36 and 13:04:05 UTC. LHASA directory listing: 04:59:47, 05:58:37 and 13:04:06 UTC. The first and second LHASA listings were 59 minutes apart, so the one-hour requirement is met by the comparison with the third listing (7 h 4 min after the first).

Source	What changed between polls	How new / changed records are detected	Revisions and deletions

Open-Meteo forecast	Polls 1→2 (70 min): only the 15-minute `current` block (time 04:45 → 05:45; precipitation changed in 4 of 10 cities). All 2,400 hourly values were identical and no hour was added or removed.	No ETag / Last-Modified. Stable key = (grid cell, hour). Watermark = last past hour stored per cell; content hash per row detects revised values.	Past hours are recomputed when a new model run is published, so each poll re-requests `past_days=2` and upserts rows whose hash changed.
NASA LHASA	Listings 1→2: a new file appeared, `20261106T0000.tif` (Last-Modified 05:07:25 UTC). Listing 3: no further file. Nothing was removed.	Watermark = newest issue time in the file names; ETag and Last-Modified confirm whether a known file was republished.	Files can be missing for days (17 gaps since 2025, up to 9 days). The listing of the last 7 days is re-checked each run.

Third poll after a new model run (13:04:05 UTC, 7 hours after the second). The Open-Meteo forecast revised **423 precipitation values and 654 soil-moisture values** out of 2,400 hourly rows, with no hours added or removed. The revisions reached hours that were already in the past at the previous poll: for Pereira, 01:00 UTC went from 1.3 mm to 0.4 mm and 03:00 UTC from 0.2 mm to 1.8 mm. Past values are therefore **not final**, which is why the contract re-reads `past_days=2` and upserts by (cell, hour) when the content hash changes. The LHASA listing showed no new file at that time (next file expected at about 05:07 UTC on 8 November).

Implication for the watermark design. In both sources the watermark is a data timestamp (last hour, last issue time), never the polling time, and every run re-reads a small overlap window. Because writes are upserts on stable keys, re-reading the overlap is safe and catches late updates.

5. Source Profiling

The notebook `notebooks/01_source_profiling.ipynb` profiles each source with more than 200 records: the full catalog (11,033 events), 7,310 daily records for 10 cities plus 1,605 daily records around the 107 Colombian landslides, 3 × 2,400 hourly forecast records, and 2 × 722,571 valid LHASA pixels plus the listing of 1,077 files.

Check	NASA catalog	Open-Meteo historical	Open-Meteo forecast	NASA LHASA
Nulls	event_time 100 %, country_name 14.2 %, fatality_count 12.6 %, injury_count 51.4 %; coordinates and dates 0 %	0 % in all fields	0 % in all fields	77 % of pixels NaN (static mask, identical on both days)
Value ranges	fatalities 0–5,000 (median 0); lat −46.8…72.6	0–175.6 mm/day; no negatives	0–12.1 mm/h; soil moisture 0.283–0.529 m^3/m^3 (three polls)	6×10^{-5} – 0.95; p99 0.31 (Nov 5) and 0.39 (Nov 6)
Duplicated ids	0 duplicated event_id; 3 probable duplicate events (Section 5.2)	0 duplicated (cell, date)	0 duplicated (cell, hour)	File names unique
Timestamp format / zone	`MM/DD/YYYY hh:mm:ss AM/PM`, local time, no zone; 50.1 % at 00:00	ISO date in the requested zone (UTC for the 10-city sample, America/Bogota for the event windows and the contract)	ISO without offset, UTC as requested	`YYYYMMDDTHHMM` in file name, UTC
Coordinates	WGS84 decimal degrees, no invalid or (0,0) values	Returned grid cell 2.2–12.0 km from the requested point	Same grid as historical	EPSG:4326, but stored south-up
Units	Accuracy as text (“5km”, “exact”, “unknown”)	mm, h (from `daily_units`)	mm, m^3/m^3 (from `hourly_units`)	Dimensionless 0–1

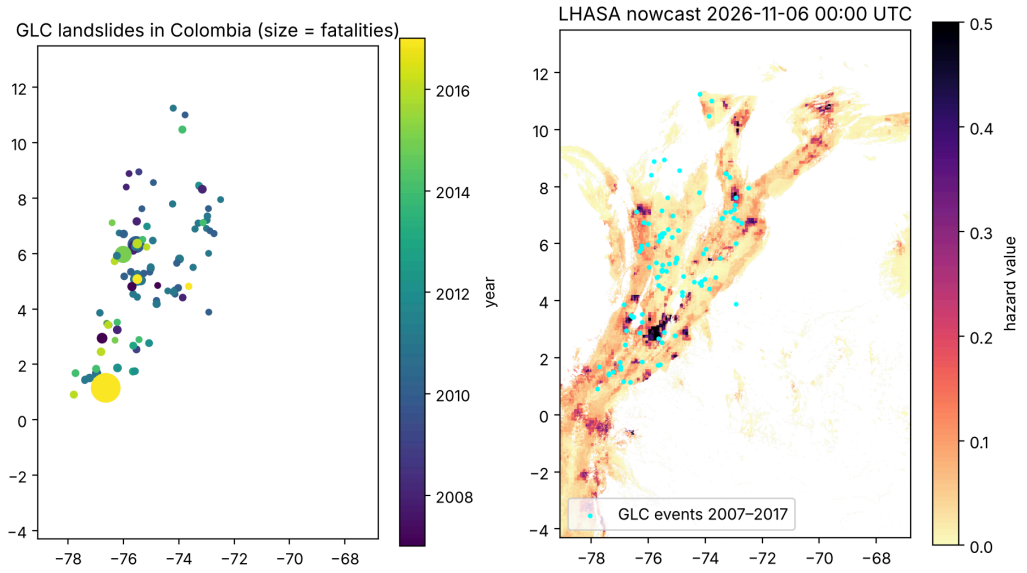


Figure 1. Left: NASA catalog landslides in Colombia, 2007–2017 (colour = year, size = fatalities). Right: LHASA nowcast of 2026-11-06 for the Colombia window with the same events (cyan).

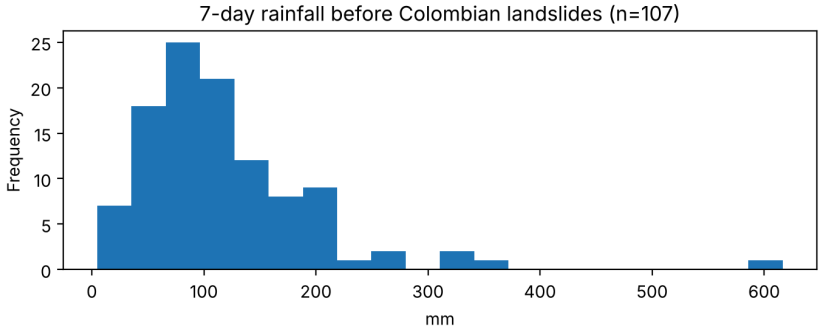


Figure 2. Seven-day rainfall (Open-Meteo ERA5) before each of the 107 Colombian landslides. Median 101.5 mm, quartiles 66.8 and 143.9 mm.

5.1 Data-quality problems (with example records)

#	Problem	Example record	Handling
1	Country missing for Colombian events, so a country filter loses 6 of 107 events. The record also dates Mocoa on 25 March, while the disaster happened on the night of 31 March–1 April 2017.	GLC 10790 “Deadly Mocoa Landslide”, country_name empty, 03/25/2017	Spatial filter (bbox + boundary); date discrepancies documented, not corrected
2	Time embedded in `event_date`; `event_time` is 100 % empty; 50.1 % of events at exactly 00:00 (unknown time); no time zone	GLC 562 “Medellin” 05/31/2008 12:00:00 AM vs. GLC 2784 12/05/2010 02:30:00 PM	Local → UTC (+5 h); `time_known = false` when 00:00
3	Location accuracy is free text with units and “unknown”	GLC 7402 “bus route from Neiva to San Vicente”: accuracy “unknown”	Convert to km; unknown → NULL; weight matches by accuracy
4	Probable duplicate records of one landslide with conflicting fatalities	GLC 3352 “El Tambo, Nariño” (0 deaths) and 3353 “San Antonio de Chuza, El Tambo” (5 deaths), same date (00:00 vs. 08:00) and 4.4 km apart	Grouped in `event_group`, not deleted
5	Open-Meteo snaps coordinates to its grid; the returned point differs from the requested one	Pasto requested (1.214, −77.281), returned (1.3005, −77.3460): 12.0 km	Store the returned cell; join on cells, not on points

6	Same field name, different meaning: in the forecast `rain` excludes showers (761 of 2,400 hours differ; totals 1,027.1 mm vs. 342.8 mm); in the archive both are equal	Manizales 2026-11-06 07:00 UTC: precipitation 3.4 mm, rain 0.1 mm	Always use `precipitation`
7	LHASA declares nodata = −9999 but uses NaN; raster stored south-up	`20261105T0000.tif`: 0 cells = −9999, 2,430,165 NaN in the window	Treat NaN as missing; flip before saving
8	LHASA publishes irregularly: changing issue times (00–21 UTC), 17 gaps (41 missing days), ~29 h delay	Gap 2025-11-23 → 2025-12-03 (9 days)	Missing files are not errors; overlap window of 7 days
9	LHASA has no value at 8 of 10 city centres (static mask on flat terrain)	Bogotá, Mocoa, Cali: NaN	Aggregate over 0.25° cells
10	“Rainfall-triggered” events with almost no rain in the reanalysis (coarse grid, imprecise location or date)	GLC 4121 “Samaná, Caldas” (downpour, accuracy 25 km): 4.9 mm in 7 days	Flag as low-confidence pairs in validation

5.2 Overlap between sources: candidate pairs

Within the catalog (same landslide reported twice; events within 3 days and 30 km). The first three are strong duplicates; the others are probably different slides of the same storm.

Pair (GLC ids)	Δ days	Distance	What differs
3353 / 3352 — El Tambo, Nariño (Apr 2011)	0	4.4 km	Time (00:00 vs. 08:00), position, fatalities 5 vs. 0, title
9226 / 9170 — Medellín–Quibdó road (Jun 2016)	0	15.1 km	Hour (15:00 vs. 14:00), position, fatalities 5 vs. 8
2331 / 2356 — Málaga, Santander (Aug 2010)	3	1.3 km	Date, title detail; same accuracy (5 km)
3359 / 3360 — Manizales (Apr 2011)	0	15.5 km	Fatalities 0 vs. 20; likely two slides
2736 / 2737 / 2738 — Caldas municipalities (20 Nov 2010)	0	14–25 km	Different places, same storm → candidate event group

Across sources (the same storm seen by the inventory and by the rainfall reanalysis). The GLC point is matched to the Open-Meteo grid cell on the local event date. What differs: position (point vs. cell centre, 2.1–6.6 km here), time (local date with unknown hour vs. a daily aggregate), severity (fatalities/size vs. millimetres) and identifier (`event_id` vs. cell + date).

GLC event	Date	Grid cell	Offset	Rain that day	Rain 7 days	Deaths
2507 Morro Azul, near Anserma	2010-09-27	5.167, −75.954	2.1 km	74.9 mm	176.6 mm	4
2720 outside Quinchía, Risaralda	2010-11-13	5.308, −75.782	6.6 km	71.2 mm	320.3 mm	4
5957 Palmira, Valle del Cauca	2014-03-13	3.550, −76.216	3.7 km	61.2 mm	209.5 mm	2
8908 road to the sea, near Cali	2016-04-12	3.480, −76.664	6.0 km	50.4 mm	179.9 mm	0
3311 Avenida Centenario, Manizales	2011-03-28	5.097, −75.520	4.5 km	44.6 mm	342.9 mm	0

LHASA vs. catalog: there is no temporal overlap (the catalog ends in 2017; the LHASA NRT archive starts in 2025), so they can only be related spatially (Figure 1). This limits the validation of the LHASA component.

6. Canonical Mapping and Data Model

The full table is in `workshop_1/mapping.md`. Every source record becomes one `event` row; hazard-specific attributes go to domain tables.

Canonical field	NASA catalog	Open-Meteo historical	Open-Meteo forecast	NASA LHASA
event_id	`nasa_glc:<event_id>`	`…:<lat>_<lon>_<date>`	`…:<lat>_<lon>_<hour>`	`…:<issue>_<cell>`
event_type	landslide	rainfall_daily	rainfall_hourly	landslide_nowcast
timestamp (UTC)	event_date (local, +5 h)	daily.time, local day start	hourly.time (UTC)	issue time from file name
latitude, longitude	latitude, longitude	returned grid lat/lon	returned grid lat/lon	0.25° cell centre
severity (unit)	landslide_size → 1–5	precipitation_sum (mm/day)	precipitation (mm/h)	max pixel value in cell (0–1)
raw_reference	source_link + raw file	request URL + raw file	request URL + raw file	file URL + ETag

Domain tables. `landslide` (catalog attributes: accuracy in km, time_known, category, trigger, size, fatalities, injuries, country, admin division, source); `rainfall_observation` (period, precipitation, precipitation hours, soil moisture, is_forecast, dataset); `lhasa_nowcast` (issue time, cell, max/mean/p90 hazard, valid pixels, ETag); `analysis_cell` (static 0.25° grid used to join everything).

Normalization rules. Timestamps in UTC (Colombia is UTC−05:00 all year); coordinates WGS84 decimal degrees; precipitation in mm from `precipitation`, never `rain`; location accuracy in km; severity always with `severity_unit`. Records with missing ids or timestamps, out-of-range coordinates, negative rainfall or LHASA values outside [0, 1] go to `rejected_record`.

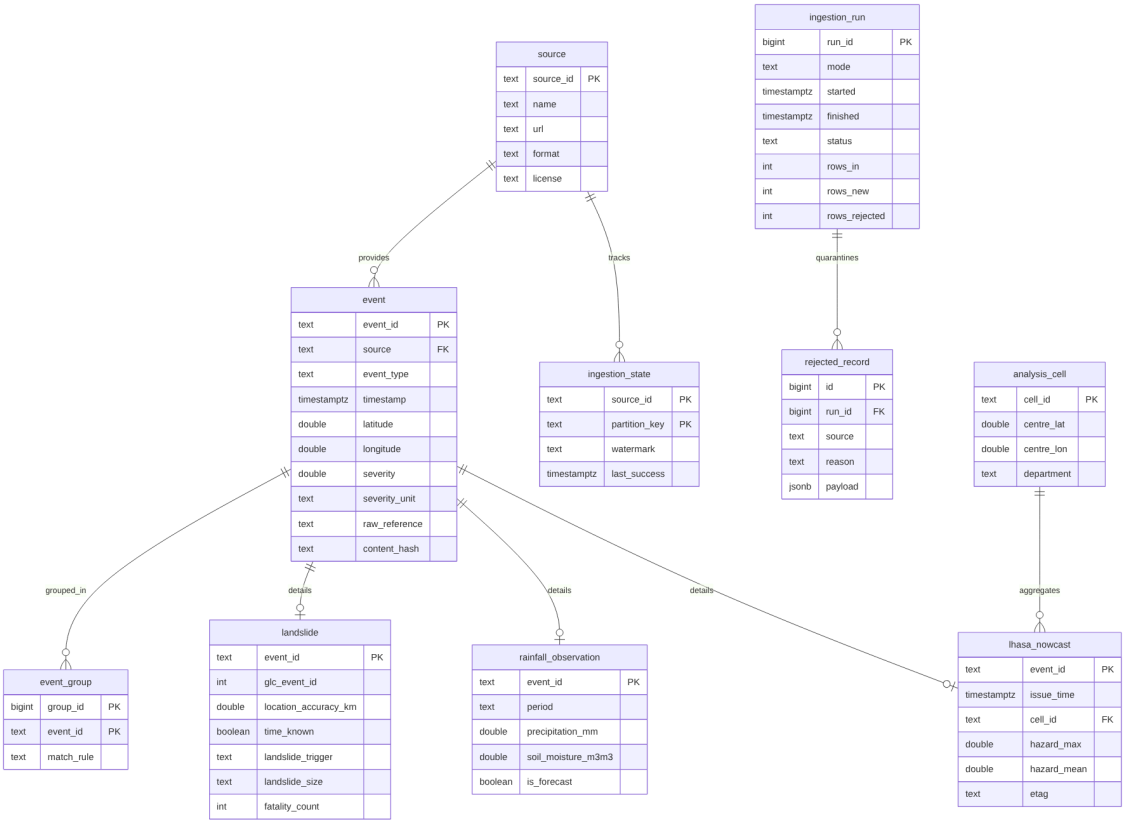


Figure 3. First draft of the entity-relationship diagram (common course tables plus domain tables).

7. Ingestion Contract (draft)

	NASA catalog	Open-Meteo historical	Open-Meteo forecast	NASA LHASA
Historical strategy	One download, no pagination	Cells × year partitions under the weight limits	—	Optional backfill since 2025 (~1.5 MB per file for Colombia)
Incremental strategy	Monthly ETag check; reload only if it changes	Monthly extension per cell	Hourly poll (minute 10)	Daily poll at 06:30 UTC
Watermark	File ETag	Last date per cell	Last past hour per cell	Last issue time processed
Overlap window	Whole file	10 days	`past_days=2`	7 days of listing
Duplicate key	event_id	(cell, date)	(cell, hour)	(issue time, cell)

Correctness after a failed run. (1) Each response is first saved unchanged in `data/raw/` as `source_YYYYMMDD_HHMMSS.ext`, so it can be re-processed without downloading again; (2) each partition is loaded in a single database transaction using upserts on the duplicate key; (3) the watermark in `ingestion_state` is advanced in that same transaction, only after the data commit. If a run stops halfway, the next run restarts from the last committed watermark, and re-inserting the overlap produces no duplicates. Every run is recorded in `ingestion_run` with its counts and status.

8. Functional User

Role. Risk analyst in a departmental disaster-risk-management office of an Andean department (for example Caldas or Antioquia).

Profile. Engineer or geoscientist; comfortable with maps and charts, not a programmer; prepares a daily situation briefing and coordinates with municipalities and road authorities.

Decisions supported. Which municipalities and road corridors to prioritise for inspection or follow-up calls today; whether rainfall in a region is unusual for the season; whether the data behind the briefing are fresh and complete.

Why the dashboard helps. It replaces checking several websites with one view that combines current and recent rainfall, the latest LHASA nowcast and the record of past landslides, and it states clearly how fresh each source is.

9. User Stories (MoSCoW)

#	Priority	Type	User story
US1	Must	Monitoring	As a risk analyst, I want to see the 24-hour and 7-day rainfall per region on a map so that I can tell where conditions changed since yesterday.
US2	Must	Monitoring	As a risk analyst, I want to see the latest LHASA nowcast per region with its issue time so that I know which areas the satellite model flags.
US3	Must	Analysis	As a risk analyst, I want to compare the current 7-day rainfall with the historical distribution for the same cell and month so that I can judge whether it is unusual.
US4	Must	Trust	As a risk analyst, I want to see when each source was last updated and the age of its watermark so that I know whether the data are fresh.
US5	Should	Analysis	As a risk analyst, I want to see past landslides (2007–2017) near a region together with the rainfall that preceded them so that I understand what conditions were dangerous before.
US6	Should	Trust	As a risk analyst, I want to see how many records were rejected and how many LHASA days are missing so that I know how complete the picture is.
US7	Could	Monitoring	As a risk analyst, I want to export the regions with a HIGH indicator as CSV so that I can attach them to my daily briefing.

US8	Won't (this semester)	Monitoring	As a risk analyst, I want to receive automatic SMS or email alerts so that I am notified without opening the dashboard.

10. Indicator Proposal

Unit of analysis. One 0.25° analysis cell per day.

Inputs. (a) R7: 7-day antecedent precipitation (Open-Meteo); (b) P: percentile of R7 within the 2007–present distribution of the same cell and calendar month; (c) L: maximum LHASA value in the cell from the latest nowcast; (d) D: number of catalog landslides within 25 km in 2007–2017 (context only).

Rule (parameterized, preliminary). Using the rainfall observed before the 107 Colombian landslides (quartiles 66.8 / 101.5 / 143.9 mm in 7 days):

- **HIGH** if R7 ≥ T_{high} (101.5 mm, the median before past landslides) **and** P ≥ 90, or if L ≥ L_{high} (to be calibrated; the 99th percentile over the Colombia bounding box was 0.31–0.39 on the two days sampled).
- **MEDIUM** if R7 ≥ T_{med} (66.8 mm, the first quartile) or P ≥ 80.
- **LOW** otherwise.

The output scale is LOW / MEDIUM / HIGH, with the numeric inputs shown next to it. All thresholds live in configuration so that a sensitivity analysis can be done in Workshop #4.

Validation against historical events. Back-test 2007–2017 with ERA5 rainfall: compute the rainfall part of the indicator for every cell and day, then measure the hit rate (share of the 107 landslides that occurred on MEDIUM or HIGH days) and the false-alarm rate (share of all cell-days with MEDIUM or HIGH and no landslide). Events with poor location accuracy or very low reanalysis rainfall (finding 10) are reported separately. The LHASA component cannot be back-tested with the catalog because the two do not overlap in time; this limitation is stated in the dashboard. The indicator is a monitoring and correlation aid, **not a prediction**: the catalog is compiled from news reports and is incomplete.

11. Risks and Mitigation

Risk	Evidence	Mitigation
Source outage or moved URL	The LHASA URL in the assignment is unreachable (HTTP 502); the catalog's old API returns 404	URLs in `sources.yaml`; source-status log; professor informed; retries with exponential backoff; dashboard shows staleness
Rate limits	Open-Meteo weight per request (e.g. 156.6 for the 2-year sample)	Partition by cells × year; read `X-RateLimit-Remaining`; stop before the daily limit and resume
Schema or format change	Catalog moved from an API to a static file; LHASA changed issue times and cadence in 2025	Validation on every run (expected columns and units); unknown records quarantined; recorded fixtures in tests
License and attribution	Open-Meteo CC BY 4.0, free tier non-commercial only	Non-commercial academic use; attribution in README, dashboard footer and report
Data volume	LHASA ~140 MB per global file; 5.7 M rainfall rows for a 0.25° grid	Range requests for the Colombia window only; aggregation to cells; cache raw files and never re-download the archive
Ambiguous identifiers and duplicates	3 probable duplicate events in the catalog; points snapped to grid cells	Deterministic ids; duplicates grouped in `event_group` with a documented rule; join on cells
Weak validation basis	Catalog ends in 2017 and is news-based; no overlap with LHASA	State the limitation; optional extra inventory (SGC SIMMA) only if programmatic access is demonstrated
Unstable network	Intermittent connection resets on the first attempts	Timeouts, retries and idempotent upserts

12. Architecture Overview

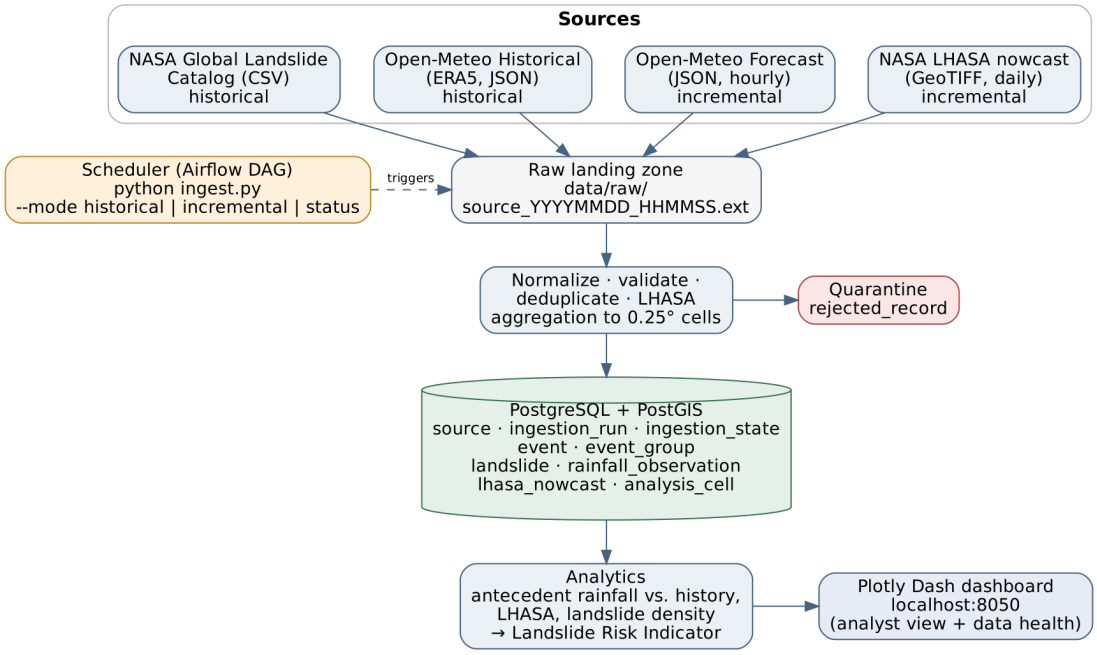


Figure 4. High-level architecture following the course reference architecture, adapted to landslides.

A scheduler (Airflow DAG) runs `python ingest.py` in historical, incremental and status modes. Raw responses land unchanged in `data/raw/`; normalization validates, deduplicates and aggregates LHASA to analysis cells; valid rows go to PostgreSQL/PostGIS and invalid ones to quarantine; the analytics layer computes the indicator, and the Plotly Dash dashboard (localhost:8050) presents it to the analyst with a data-health panel.

13. Repository

The repository follows the required structure: `src/` (sources, normalization, storage), `ingest.py` (stub with the three modes), `config/sources.yaml` (one entry per source with URL, format, schedule, watermark, license and attribution), `data/raw/` (git-ignored), `sql/schema.sql` (draft DDL of the ER diagram), `dashboard/` (placeholder with the planned views), `tests/` (three passing tests and recorded fixtures), `notebooks/01_source_profiling.ipynb` (executed), `docs/` (source-status log, ER and architecture diagrams), `workshop_1/` (samples, mapping, download script), `.env.example`, `.gitignore` and `README.md`. No secrets are committed.

14. References and Attribution

- Kirschbaum, D.B., Adler, R., Hong, Y., Hill, S., Lerner-Lam, A. (2010). A global landslide catalog for hazard applications: method, results, and limitations. *Natural Hazards*, 52, 561–575.
- Kirschbaum, D.B., Stanley, T., Zhou, Y. (2015). Spatial and temporal analysis of a global landslide catalog. *Geomorphology*, 249, 4–15.
- Kirschbaum, D., Stanley, T. (2018). Satellite-based assessment of rainfall-triggered landslide hazard for situational awareness. *Earth's Future*, 6, 505–523.
- Stanley, T.A., Kirschbaum, D.B., et al. (2020). Building a landslide hazard indicator with machine learning and land surface models. *Environmental Modelling & Software*, 129, 104692. https://doi.org/10.1016/j.envsoft.2020.104692
- NASA Global Landslide Catalog Export: https://data.nasa.gov/dataset/global-landslide-catalog-export
- NASA LHASA nowcast (NRT): https://portal.nccs.nasa.gov/datashare/landslides/nrt/ (linked from https://gpm.nasa.gov/applications/landslides)
- Weather data by Open-Meteo.com, CC BY 4.0: https://open-meteo.com — ERA5 reanalysis, Copernicus Climate Change Service / ECMWF.
- Source changes detected during this workshop: `docs/source_status_log.md`.